

Final Exam (Summer Course)

Q1: True or False?

- 1- The fragment shader determines the position of the vertices. **False**
- 2- Two translation matrices, T_1 and T_2 can be combined into another translation matrix T by adding them, i.e. $T = T_1 + T_2$. **False**
- 3- For a rotation matrix R and a translation matrix T , $RT = TR$, even though matrix multiplication is not generally commutative. **False**
- 4- The Phong reflectance model does not consider the geometry of the object being shaded. **False**
- 5- The Phong reflectance model calculates the red, green, and color channels independently. **True**
- 6- For mipmapping, the use of resolution levels requires n times the amount of memory. **False**
- 7- (x, y, z, w) can be used to represent either a 3D point or a 3D vector. **True**
- 8- The product of several rotation matrices is itself a rotation. **True**
- 9- Rasterization occurs after vertex transformation in the graphic. **True**
- 10- The OpenGL shading language statement "float foo = 1" is incorrect one. **True**
- 11- Reflection of a 3D object around a given axis is equivalent to its rotation by 90° around that axis. **False**
- 12- Translation and Rotation are considered to be rigid-body transformations, which move objects without any deformation or distortion. **True**
- 13- If the rotation matrix R rotates a point by the angle θ , then the inverse of R rotates the point by $-\theta$. **True**
- 14- In orthographic projection, objects that are further away from the viewer appear smaller. **False**
- 15- The term ambient means that light is perfectly reflected in a mirror-like way from the light source to the viewer. **False**
- 16- The OpenGL Shading Language (GLSL) enables programmers to replace the OpenGL fixed-function with vertex and fragment shaders. **True**
- 17- OpenGL runs only on Microsoft operating system. **False**

Q2: Choose:

- 18- Given $U = (1, 2, -1)$ and $V = (0, -1, 3)$, the cross product $U \times V$ equals.....

A) (0, -2, 3)	B) (5, -3, -1)	C) (1, 1, 2)	D) (5, 3, -2)
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- 19-light component comes from one direction and once it hits a surface, it is scattered.

A) Diffuse	B) Specular	C) Ambient	D) Spot
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20- light comes from one direction and tends to bounce off the surface in a preferred direction.

A) Diffuse	B) Specular	C) Ambient	D) Spot
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21-light is so scattered; it is impossible to tell what its original.

A) Diffuse	B) Specular	C) Ambient	D) Spot
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22-projection preserves parallel lines.

A) Orthographic	B) Perspective	C) Frustum	D) Symmetric
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23- The dot product of any two perpendicular vectors equals.....

A) 0	B) 1	C) (0, 0, 0)	D) (1, 1, 1)
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24- If the GL_SPOT_CUTOFF is, spotlight effect equals one.

A) 0	B) 45	C) 90	D) 180
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25-is the OpenGL technique that takes a texture map and creates a hierarchy of coarser and coarser versions of it.

A) FOV	B) Culling	C) Antialiasing	D) Mipmapping
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26- The process of backface..... makes rendering objects quicker and more efficient by reducing the number of polygons for the program to draw.

A) FOV	B) Culling	C) Antialiasing	D) Mipmapping
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27- Write down the Modelview matrix after executing the following OpenGL code segment.

`glMatrixMode(GL_MODELVIEW);`

`glLoadIdentity();`

`glScalef (-2.0,1.5, -3.0);`

A) $\begin{bmatrix} 0 & -1 & 0 & -2 \\ 1 & 0 & 0 & 1.5 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	B) $\begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 1.5 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	C) $\begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 1.5 & 0 & 0 \\ 0 & 0 & -3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	D) $\begin{bmatrix} 0 & 0 & -2 & 0 \\ 0 & 1.5 & 0 & 0 \\ -3 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
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28- Write down the Modelview matrix after executing the following OpenGL code segment.

```
glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glRotatef(90.0, 0, 0, 1.0);
```

A) $\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	B) $\begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	C) $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	D) $\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
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29- Write down the Modelview matrix after executing the following OpenGL code segment.

```
glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glRotatef(-90.0, 0, 0, 1.0);
glRotatef(180.0, 0, 0, 1.0);
```

A) $\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	B) $\begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	C) $\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	D) $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
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30- Write down the Modelview matrix after executing the following OpenGL code segment.

```
glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glPushMatrix();
glRotatef(90.0, 0, 0, 1.0);
glScalef (-1.0,2.0, -1.0);
glPopMatrix();
```

A) $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	B) $\begin{bmatrix} 0 & 2 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	C) $\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	D) $\begin{bmatrix} 0 & 1 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
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31- How many vertices expected in one triangle has 100 fragments is sent as input drawing by the hardware?

A) 1	B) 3	C) 100	D) 300
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32- How many vertices expected in one triangle has 100 fragments is sent as input drawing by the hardware?

A) 1	B) 3	C) 100	D) 300
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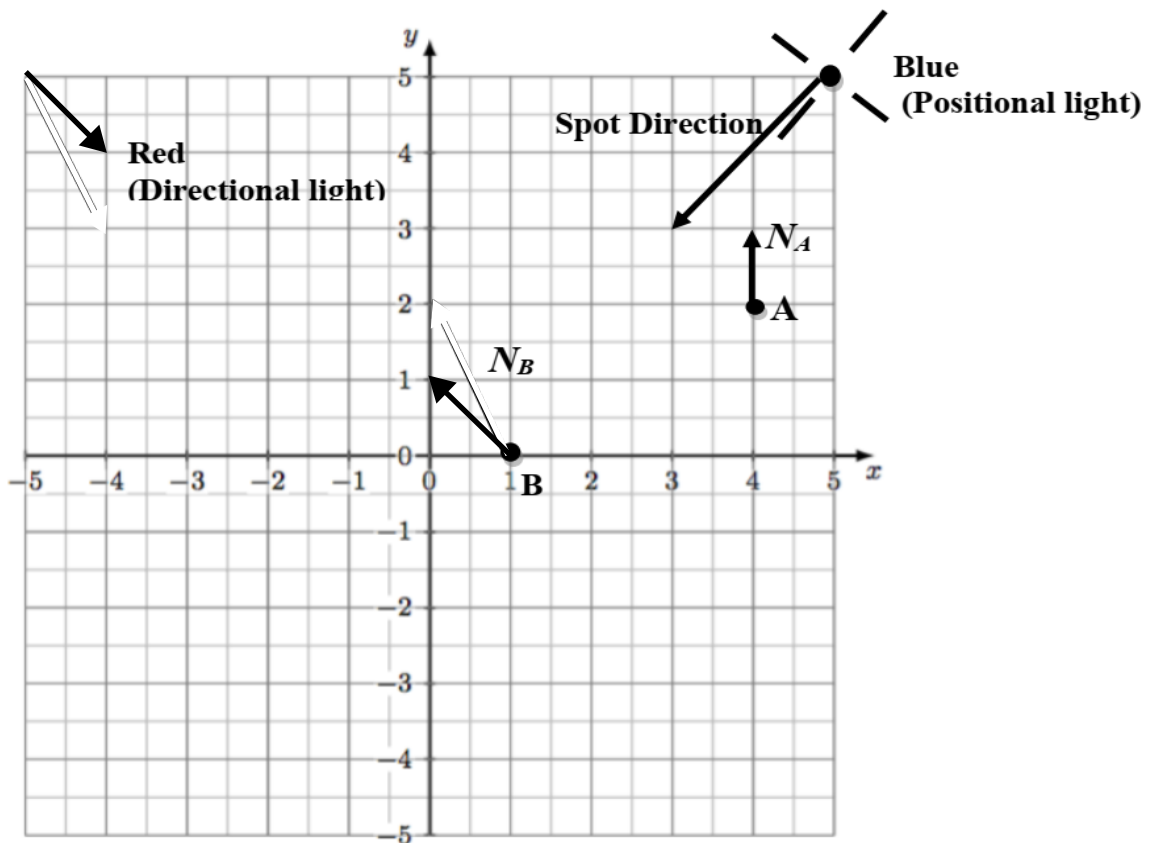
33- How many vertices expected in one triangle has 100 fragments is sent as input drawing by the hardware?

A) 1	B) 3	C) 100	D) 300
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34- The RGB of A yellow color is.....

A) (1.0,1.0,0.0)	B) (1.0,0.0,1.0)	C) (0.0,1.0,1.0)	D) (1.0,1.0,1.0)
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The figure below contains two light sources and two points in 2D. The left light source emits red diffuse light = (1.0, 0.1, 0.1) and its spotlight effect=1. The right light source emits blue diffuse light = (0.1, 0.2, 1.0), spot cutoff = 20, and spot exponent=2.0. The normal vectors are shown in the figure. Assume the constant attenuation=0, linear attenuation =1.0, quadratic attenuation=0, the material ambient = (0, 0, 0), the material emission=(0.1,0.1,0.1), the diffuse components of the material = (0.6,0.8,0.4), and the specular components of the material = (0, 0, 0) for the two points. Using the Phong reflection model, answer the following questions:



35- The normalized normal vector N_A =

A) (0,1)	B) (1,0)	C) (0,-2)	D) (-2,0)
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36- The normalized normal vector N_B =

A) (-1,1)	B) $(\frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$	C) $(\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}})$	D) (1,-1)
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37- The diffuse intensity for the point **A** for the blue light source =

A) $\frac{1}{2\sqrt{10}}$	B) $\frac{3}{\sqrt{5}}$	C) $\frac{2}{\sqrt{5}}$	D) $\frac{3}{\sqrt{10}}$
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38- The diffuse intensity for the point **B** for the blue light source = $\sqrt{82} / 82$ OR $1 / \sqrt{82}$

A) $\frac{-4}{\sqrt{12}}$	B) $\frac{1}{\sqrt{12}}$	C) $\frac{1}{\sqrt{41}}$	D) $\frac{-1}{\sqrt{41}}$
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39- The attenuation factor for the point **A** for the blue light source =

A) $\frac{1}{\sqrt{10}}$	B) 1	C) $\frac{1}{2\sqrt{41}}$	D) $\frac{1}{2\sqrt{10}}$
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40- The attenuation factor for the point **A** for the red light source =

A) $\frac{1}{\sqrt{10}}$	B) $\frac{1}{2\sqrt{41}}$	C) 1	D) $\frac{1}{2\sqrt{10}}$
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